

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – COMPARITIVE ANALYSIS

Course Code:	ITA03	Course Title:	Mobile Computing
CO Assessed:	CO4	Assessment:	AT1 - COMPARITIVE ANALYSIS
Weightage:	30 % of CIA	Total Marks:	100
CO4 Statement:	Identify the components and structure of mobile application for Android and Windows OS-based devices.		
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Section / Batch:	Slot D	Date:	18-08-2026

Code of Conduct

I S Sandhiya(192321015) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Duration: 60 minutes
- Number of questions : 01

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure A - QUESTIONS

A company is evaluating whether a legacy application should continue running on Symbian OS or be redesigned for iOS. Compare the two platforms in terms of kernel architecture, application frameworks, security, memory management, and hardware support. Determine whether migration to iOS would provide significant technical advantages.

Symbian OS vs iOS

Introduction

Symbian OS and iOS are mobile operating systems designed for different generations of smartphones. Symbian was widely used in older mobile phones, while iOS is designed for modern Apple devices. Here, we compare Symbian OS and iOS under five simple headings: kernel architecture, application frameworks, security, memory management, and hardware support.

1. Kernel Architecture (How the operating system works)

- **Symbian OS:** Uses a microkernel-based architecture. Only essential services run in the kernel, while other services are handled separately. This makes the system lightweight and suitable for devices with limited resources.
- **iOS:** Uses a hybrid XNU kernel, which combines features of microkernel and monolithic designs. It provides strong process management, memory protection, device management, and security.

Simple takeaway: Symbian was designed for resource-limited mobile phones, while iOS provides a more modern and powerful kernel architecture with better support for advanced smartphone features.

2. Application Frameworks (How apps are developed)

- **Symbian OS:** Uses frameworks such as Avkon and Symbian C++ for developing applications. These frameworks were designed mainly for older mobile devices and require more complex programming.
- **iOS:** Uses modern frameworks such as UIKit, SwiftUI, Foundation, and Core Foundation. Developers can use Swift and Objective-C to create applications.

Simple takeaway: iOS provides a more modern and developer-friendly application framework. It also offers better tools and APIs for building advanced applications.

3. Security (How the system protects data)

- **Symbian OS:** Provides security features such as application signing, permissions, and platform security. However, its security model was designed for an older mobile environment.
- **iOS:** Provides strong security through application sandboxing, code signing, permission controls, secure boot, encryption, and hardware-based security features.

4. Memory Management (How memory is used and controlled)

- **Symbian OS:** Uses manual and system-level memory management techniques. Developers had to carefully manage memory because older Symbian devices had limited RAM.
- **iOS:** Uses Automatic Reference Counting (ARC) to automatically manage the lifetime of objects. iOS also provides advanced memory-management mechanisms to handle limited resources efficiently.

Simple takeaway: iOS provides easier and more predictable memory management for modern application development, while Symbian requires more careful resource management.

5. Hardware Support (How the OS works with devices)

- **Symbian OS:** Supported hardware from different mobile manufacturers such as Nokia, Sony Ericsson, and Samsung. However, hardware capabilities were generally limited compared with modern smartphones.
- **iOS:** Is designed specifically for Apple hardware such as the iPhone and iPad. This close integration allows iOS to optimize performance, graphics, sensors, camera, processor, and other hardware components.

Simple takeaway: Symbian supported a wider range of older mobile devices, but iOS provides much tighter hardware-software integration and better performance on modern devices.

Visual Comparison

Feature	Symbian OS	iOS
Kernel Architecture	Microkernel-based	Hybrid XNU
Application Framework	Symbian C++, Avkon	Swift, Objective-C, UIKit, SwiftUI
Security	Platform security, permissions, signing	Sandboxing, encryption, code signing, Secure Boot
Memory Management	Manual/system-level management	Automatic Reference Counting (ARC)
Hardware Support	Multiple older manufacturers	Optimized for Apple hardware

Development	More complex	Modern and developer-friendly
Performance	Suitable for older devices	Better optimized for modern hardware